# Corey Lynn Murphey

corey.murphey@colorado.edu
Website | ORCID | Github

## EDUCATION

**Ph.D.** — *In Progress*
**University of Colorado – Boulder** **8/2021 – present**
*Department of Computer Science*
Advisors: Elizabeth Bradley, Allison Hilger, and Jed Brown
Focus: Numerical and Scientific Computing

**M.S.**
**University of Colorado – Boulder** **8/2021 - 5/2024**
*Department of Computer Science*
Advisors: Elizabeth Bradley and Jed Brown
Focus: Numerical and Scientific Computing

**M.S.**
**Stanford University** **4/2012 – 4/2014**
*Department of Mechanical Engineering*
Advisor: Reginald Mitchell
Focus: Energy Systems; Breadth: Biomechanics and Manufacturing

**B.S.**
**Stanford University** **8/2008 – 1/2013**
*Department of Mechanical Engineering*
Advisor: Ellen Kuhl
Focus: Computational Biomechanics and Biomechanical Engineering

## RESEARCH EXPERIENCE

6/2021 – present **University of Colorado – Boulder**
*Graduate Research Assistant, Advised by Elizabeth Bradley and Allison Hilger*

Developed models of phonation and vibration-induced aerosolization.

Simulated fluid-structure interaction within the larynx.

Generated musical chord progressions using chaotic dynamics and the Tonnetz.

Analyzed Indian ocean sediment core data using Information Theory.

9/2012 – 6/2013 **Stanford University, Hearing Dynamics**
*Research Assistant, Advised by Sunil Puria*

Built a model of Békésy's pendulum to demonstrate hair cell dynamics.

Developed a computational model of Békésy's analogy for the inner ear.

5/2010 – 8/2012 **Stanford University, Living Matter Laboratory**
*Research Assistant, Advised by Ellen Kuhl*

Modeled electrochemical conductive pathways of the heart.

Generated electrocardiogram plots of simulated cardiac pacing.

Developed patient-specific models of implant-induced skin growth.

Worked with graduate students to create a model of red blood cell division.

Designed a continuum growth model of the vocal folds and vocal polyps.

## PROFESSIONAL EXPERIENCE

10/2018 - 7/2021 **Bolder Industries**, Boulder, CO
*R&D Engineer, IP Manager, and Chief of Staff*

Modeled tire pyrolysis reaction kinetics to anticipate pyrolytic outputs.

Implemented an intellectual property strategy to protect Bolder Industries' IP.

5/2018 – 8/2018 **Caban Systems**, San Mateo, CA
*Thermal Engineer, Consultant*

Modeled heat emitted by batteries inside an energy-storage cabinet.

Calculated cooling required for peak operation of batteries in the cabinet.

3/2017 – 4/2018 **Run8 Patent Group**, San Francisco, CA
*Patent Agent*

Drafted and prosecuted over 50 patent applications.

Managed domestic and foreign patent portfolios.

6/2015 – 3/2017 **Nebia**, San Francisco, CA
*R&D Engineer and Engineering Project Manager*

Simulated heat-transfer from droplets emitted from showerhead nozzles.

Modeled internal fluid pathways optimized for nozzle performance.

Drafted and maintained product requirements documents.

4/2014 – 4/2015 **Schox Patent Group**, San Francisco, CA
*Patent Agent*

Drafted patent applications and managed portfolios of startups.

Maintained a patent portfolio with over 50 distinct projects.

6/2013 – 9/2013 **Benvenue Medical Inc.**, Santa Clara, CA
*R&D Engineering Intern*

Established testing protocols for vertebral augmentation implants.

Developed surgical protocols for mixing and dispensing bone cement.

## TEACHING

Spring 2026 **Numerical Computation (CSCI 3656)**
*Teaching Assistant*, University of Colorado – Boulder

Spring 2026 **Chaotic Dynamics (CSCI 4446/5446)**
*Grader*, University of Colorado – Boulder

Summer 2025 **Algorithms (CSCI 3104)**
*Teaching Assistant*, University of Colorado – Boulder

Spring 2025 **Numerical Solutions to Partial Differential Equations (CSCI 5636)**

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| | *Teaching Assistant*, University of Colorado – Boulder |
| Spring 2025 | **Chaotic Dynamics (CSCI 4446/5446)**<br>*Grader & Course Assistant*, University of Colorado – Boulder |
| Fall 2024 | **Numerical Computation (CSCI 3656)**<br>*Teaching Assistant*, University of Colorado – Boulder |
| Spring 2023 | **Chaotic Dynamics (CSCI 4446/5446)**<br>*Course Manager*, University of Colorado – Boulder |
| Fall 2013 | **Patent Law and Strategy for Engineers (ME 208)**<br>*Course Assistant*, Stanford University |
| Fall 2012 | **Engineering Dynamics (E15)**<br>*Grader*, Stanford University |

### Guest Lectures

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| Summer 2025 | "Dynamic Programming," Algorithms (CSCI 3104) |
| Spring 2025 | "Stiffness & Runge-Kutta Methods," Numerical Solutions to PDEs (CSCI 5636) |
| Fall 2024 | "To Grad School or Not Grad School," Advanced Data Structures (CSCI 2270) |

## AWARDS AND HONORS

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| 2024 | **Outstanding Departmental Service Award**<br>University of Colorado – Boulder, Computer Science Department |
| 2024 | **Poster Award - Work in Progress Research, Computer Science Research Expo**<br>University of Colorado – Boulder, Computer Science Department<br>For the work-in-progress poster entitled "Generation of Novel Chord Progressions via a Musically-Inspired Chaotic Mapping" with primary author, Zachary Atkins. |
| 2024 | **First & Second Prize - Poster Awards, Dynamics Days 2024**<br>For the posters entitled "A Dynamics-Inspired Model for Phonation-Induced Aerosolization" and "Generation of Novel Chord Progressions via a Musically-Inspired Chaotic Mapping". |
| 2023 | **D. J. Kasik (1972) Scholarship Fund Award**<br>University of Colorado – Boulder, College of Engineering and Applied Sciences |
| 2023 | **Outstanding Departmental Service Award**<br>University of Colorado – Boulder, Computer Science Department |
| 2022 | **CS Endowed Founder's Fellowship**<br>University of Colorado – Boulder, Computer Science Department |
| 2021 | **Early Career Professional Development Fellowship**<br>University of Colorado – Boulder, Computer Science Department |

## GRANTS

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| 2019 | **Colorado Advanced-Industries Early-Stage Capital and Retention Grant**<br>State of Colorado, OEDIT |
| 2010 – 2012 | **Vice Provost of Undergraduate Education (VPUE) Grant** |

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| | Stanford University |
| 2010 – 2012 | **Summer Undergraduate Research Institute (SURI) Grant**<br>Stanford University |
| 2008 | **Stanford Summer Engineering Academy (SSEA) Grant**<br>Stanford University |

## CONFERENCES

2025 Dynamics Days US 2026, Tucson, AZ.

2025 SIAM Conference on Applications of Dynamical Systems (DS25), Denver, CO.

2025 Dynamics Days US 2025, Denver, CO.

2024 Dynamics Days US 2024, Davis, CA.

2023 American Association for Aerosol Research (AAAR) $41^{st}$ Annual Conference, Portland, OR.

2023 SIAM Conference on Applications of Dynamical Systems (DS23), Portland, OR.

2023 15th International Conference on Advances in Quantitative Laryngology, Voice and Speech Research 2023, Phoenix, AZ.

2023 Dynamics Days US 2023, Virtual.

2011 ASME 2011 Summer Bioengineering Conference, Portland, OR.

2011 IUTAM Symposium on Computer Models in Biomechanics, Stanford, CA.

### Conference & Travel Grants

2026 **Dynamics Days Travel Fellowship**
*For travel to Dynamics Days US 2026.*

2025 **Conference Support Fellowship**
Department of Computer Science, University of Colorado – Boulder.

2025 **Dynamics Days Travel Fellowship**
*For travel to Dynamics Days US 2025.*

2024 **Dynamics Days Travel Fellowship**
*For travel to Dynamics Days US 2024.*

2024 **Clive Bailie Memorial Conference Support Fellowship**
Clive Bailie Memorial Fund, University of Colorado – Boulder.
*For travel to Dynamics Days US 2024.*

2023 **AAAR US 2023 Student Travel Grant**
American Association for Aerosol Research.
*For American Association for Aerosol Research (AAAR) $41^{st}$ Annual Conference.*

2023 **Conference Support Fellowship**
Department of Computer Science, University of Colorado – Boulder.
*For SIAM Conference on Applications of Dynamical Systems (DS23).*

2023 **AQL 2023 Student Registration Award**
*For 15th International Conference on Advances in Quantitative Laryngology, Voice and Speech Research 2023.*

2023 **Graduate School Domestic Travel Grant**
University of Colorado – Boulder.
*For 15th International Conference on Advances in Quantitative Laryngology, Voice and Speech Research 2023.*

## PUBLICATIONS

2025 A. Hilger, T. Stockman, **C. Murphey**, J. McCurdy, and S. Miller "The influence of glottal and respiratory factors on aerosol emission during phonation," PLOS ONE 20(12): e0321909, 2025, `https://doi.org/10.1371/journal.pone.0321909`.

### Reviewed Conference Papers

2011 **C. L. Murphey**, J. Wong, and E. Kuhl, "Computational Simulation of Biventricular Pacing in an Asymptomatic Human Heart," in SBC2011, ASME 2011 Summer Bioengineering Conference, Parts A and B, Jun. 2011, pp. 917–918, doi: 10.11105/SBC2011-53110.

2011 **C. L. Murphey**, J. Wong, and E. Kuhl, "Computational Simulation of Biventricular Pacing in a Human Heart," in Proceedings of the IUTAM Symposium on Computer Models in Biomechanics, Stanford, California, 2011.

### Presentations

#### Talks

2026 **C. L. Murphey**, A. Hilger, E. Bradley, "Fluid Atomization from a Clarinet Reed," Dynamics Days 2026, Tucson, AZ, Jan. 2026.

2025 **C. L. Murphey**, E. Bradley, "From Climate to Clarinet Reeds: Navigating Flawed Data in Computational Physics," Vassar College Physics Department, Poughkeepsie, NY, Nov. 2025. [*Invited Talk*].

2025 **C. L. Murphey**, E. Bradley, A. Hilger, "A Computational Model of Phonation-Induced Aerosolization," SIAM Conference on Applications of Dynamical Systems (DS25), Denver, CO, May 2025.

#### Posters

2025 **C. L. Murphey**, A. Hilger, E. Bradley, "A Computational Model of Phonation-Induced Aerosolization," Dynamics Days US 2025, Denver, CO, Jan. 2025.

2024 **C. L. Murphey**, A. Hilger, E. Bradley, "A Dynamics-Inspired Model for Phonation-Induced Aerosolization," University of Colorado – Boulder Computer Science Department's Research Expo Poster Session, Feb. 2024.

2024 Z. Atkins, **C. L. Murphey**, "Generation of Novel Chord Progressions via a Musically-Inspired Chaotic Mapping," University of Colorado – Boulder Computer Science Department's Research Expo Poster Session, Feb. 2024[1].

2024 **C. L. Murphey**, A. Hilger, E. Bradley, "A Dynamics-Inspired Model for Phonation-Induced Aerosolization," University of Colorado – Boulder Applied Math Department's Research Poster Session, Feb. 2024.

2024 **C. L. Murphey**, A. Hilger, E. Bradley, "A Dynamics-Inspired Model for Phonation-Induced Aerosolization," Dynamics Days US 2024, Davis, CA, Jan. 2024[2].

[1] Awarded a department award for best work-in-progress poster at the Computer Science Research Expo.
[2] Awarded best poster at Dynamics Days 2024 in Davis, CA.

2024 Z. Atkins, **C. L. Murphey**, “Generation of Novel Chord Progressions via a Musically-Inspired Chaotic Mapping,” Dynamics Days US 2024, Davis, CA, Jan. 2024[3].

2023 **C. L. Murphey**, A. Hilger, E. Bradley, “An Experimentally Validated Model of Phonation-induced Aerosolization,” American Association for Aerosol Research 41st Annual Conference (AAAR 2023), Portland, OR, Oct. 2023.

2023 **C. L. Murphey**, A. Hilger, E. Bradley, “A Dynamics-Inspired Model for Phonation-Induced Aerosolization,” SIAM Conference on Applications of Dynamical Systems (DS23), Portland, OR, May 2023.

2023 **C. L. Murphey**, A. Hilger, E. Bradley, “A Computational Model of Phonation-Induced Aerosolization,” 15th International Conference on Advances in Quantitative Laryngology, Voice and Speech Research 2023, Phoenix, AZ, Mar. 2023.

2023 **C. L. Murphey**, A. Hilger, E. Bradley, “A Dynamics-Inspired Model for Phonation-Induced Aerosolization,” University of Colorado – Boulder Applied Math Department’s Research Poster Session, Mar. 2023.

2023 **C. L. Murphey**, A. Hilger, E. Bradley, “A Dynamics-Inspired Model for Phonation-Induced Aerosolization,” Dynamics Days US 2023, Virtual, Jan. 2023.

2011 **C. L. Murphey**, J. Wong, and E. Kuhl, “Computational Simulation of Biventricular Pacing in an Asymptomatic Human Heart,” ASME Summer Bioengineering Conference, Farmington, PA, Jun. 2011.

### Patents

*Inventor*

2020 US D881,340, “Showerhead and arm,” Apr. 14, 2020.

2019 US 10,421,083, “Immersive showerhead,” Sep. 24, 2019.

2019 US D855,759, “Shower wand,” Aug. 06, 2019.

2018 US 9,931,651, “Immersive showerhead,” Apr. 03, 2018.

2018 US 9,925,545, “Immersive showerhead,” Mar. 27, 2018.

2018 US D810,233, “Shower wand and adjustable mount,” Feb. 13, 2018.

2018 US D810,234, “Showerhead and adjustable bracket,” Feb. 13, 2018.

### Miscellaneous Publications

2025 **C. L. Murphey**, “Fluctus Cloud,” APS Division of Fluid Physics, Traveling Gallery of Fluid Motion, Houston Museum of Natural Science, 2025. [*Photograph*].

2016 G. Parisi-Amon and **C. L. Murphey**, “Full Steam Ahead,” ANSYS Advantage, vol. 10, no. 1, pp. 10–12, 2016. [*Magazine Article*].

2013 [*Contributor and Editor*] J. Schox, Not So Obvious: An Introduction to Patent Law and Strategy, 3rd ed. CreateSpace Independent Publishing Platform, 2013. [*Book*].

## SERVICE

### Academic Service at the University of Colorado – Boulder

2024 – Present **Computing Advisory Board**[4]

[3]Awarded second prize for poster awards at Dynamics Days 2024 in Davis, CA.

[4]A collaboration between industry partners and the School of Computing to discuss strategic initiatives for University of Colorado computing programs and departments.

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| | *Ph.D. Student Member and Representative* |
| Fall 2024 | **Computer Science Ph.D. Application Feedback Program**<br>*Mentor* |
| Spring 2024 | **Computer Science Ph.D. Open House**<br>*Graduate Student Organizer*, Computer Science Department |
| Fall 2023 | **Computer Science Graduate Student Association (CSGSA)**<br>*CSGSA Chair*, Computer Science Department |
| Fall 2023 | **Computer Science Ph.D. Application Feedback Program**<br>*Mentor and Program Organizer*, Computer Science Department |
| Spring 2023 | **Computer Science Ph.D. Open House**<br>*Graduate Student Organizer and Panelist*, Computer Science Department |
| 2022 – 2023 | **Computer Science Graduate Committee**<br>*Ph.D. Student Representative*, Computer Science Department |
| Spring 2022 | **Summer Program for Undergraduate Research (SPUR)**<br>*Advisor to Mentors*, College of Engineering and Applied Sciences |
| Spring 2022 | **Discovery Learning Apprenticeship (DLA) Program**<br>*Mentor and Judge*, College of Engineering and Applied Sciences |
| Spring 2022 | **Computer Science Ph.D. Student Open House**<br>*Graduate Student Panelist*, Computer Science Department |

### Mentorship at the University of Colorado – Boulder[5]

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| 2024 – 2025 | Wisang Sugiarta, Ph.D. Student, Computer Science Department |
| 2022 – 2023 | Zachary Atkins, Ph.D. Student, Computer Science Department |
| 2022 – 2023 | Maria Valentini, Ph.D. Student, Computer Science Department |
| 2022 – 2023 | Aditya Pandey, M.S. Student, Computer Science Department |
| 2022 – 2023 | Armin Gholampoor, M.S. Student, Computer Science Department |

## PROFESSIONAL MEMBERSHIPS, AFFILIATIONS, & CERTIFICATIONS

### Certifications

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| 2015 – Present | United States Patent and Trademark Office, Registered Patent Agent |

### General Membership

Acoustical Society of America (ASA)

Society for Industrial and Applied Mathematics (SIAM)

The Voice Foundation

Society of Women Engineers (SWE)

American Society of Mechanical Engineers (ASME)

[5] Peer mentorship and project mentorship for fellow graduate students.

**Other Service and Affiliations**

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| 2025 – present | Boulder High School Women's Swimming : Assistant Coach |
| 2025 – present | Boulder Aquatic Masters: Coach |
| 2021 – 2024 | Westview Lutheran Church: Alto section leader |
| 2021 – 2023 | Renova New Music Ensemble: Founding Member, Webmaster, Soprano/Alto |
| 2021 – 2023 | CU – Chamber Singers: Alto 1 |
| 2018 – 2021 | St. Thomas Aquinas – Boulder: Cantor, Soprano 2 Section Leader |
| 2018 – 2020 | St. Vrain Innovation Center: Middle School Robotics Mentor |
| 2012 – 2018 | NorCal Golden Retriever Rescue : Volunteer |
| 2011 – 2012 | Stanford Women's Varsity Swimming: Team Manager |

Updated January 2026